

# IMPROVED KISSING NUMBERS IN NINETEEN AND TWENTY-ONE DIMENSIONS

ABSTRACT. We prove that the kissing numbers in 19 and 21 dimensions are at least 12268 and 30779. The best previously published lower bounds were 11948, due to Ho, and 29768, due to Cohn and Li, so the gains are 320 and 1011. Both constructions start from the odd-sign kissing configurations of Cohn and Li, delete a small block of their points, and add more new points than were deleted. In 19 dimensions we cut the 20-dimensional configuration by the hyperplane orthogonal to the indicator vector of a tetrad, project 20-dimensional sign vectors into that hyperplane, and give their tails one of two lengths. The proof is a case analysis. In 21 dimensions the new points are weighted sign vectors whose sign patterns meet all eight cosets of a  $[21, 9, 6]$  code. They are given by an explicit table of 192 integer vectors and a group of 16 sign changes, and the proof is an exact computation, reduced by that symmetry. We have also formalized both results in Lean 4 with Mathlib.

## 1. INTRODUCTION

The kissing problem in  $\mathbb{R}^n$  asks how many unit spheres can touch a central unit sphere without overlapping [2, Ch. 1, §2.2]. In terms of the points of tangency, it asks for the largest set  $X$  of unit vectors in  $\mathbb{R}^n$  with  $\langle x, y \rangle \leq 1/2$  for all distinct  $x, y \in X$ . We write  $\tau_n$  for this maximum, as in [2, Ch. 13]. It is known only for  $n \in \{1, 2, 3, 4, 8, 24\}$ ; see [1, Table 1.1] for references.

For  $17 \leq n \leq 23$  the best constructions were Leech’s cross sections of the Leech lattice minimal vectors [7] until Cohn and Li [1] improved dimensions 17 through 21. In 19, 20 and 21 dimensions their idea is short. Take the classical configuration built from a constant-weight code [8], and put an odd number of minus signs on each codeword instead of an even number. Every sign vector of a code orthogonal to the constant-weight code then has inner product at most 6 with the old points, while the kissing condition allows  $\frac{1}{2}\sqrt{8n}$ , which is larger for  $n \geq 19$ . So these sign vectors can be added, one per word of a code with minimum distance at least  $n/4$ . Cohn and Li obtained  $\tau_{19} \geq 11692$  in this way, and Ho [6] found a larger code of 1280 words for the same purpose, reaching  $\tau_{19} \geq 11948$ .

**Theorem 1.1.** *We have  $\tau_{19} \geq 12268$  and  $\tau_{21} \geq 30779$ .*

Table 1 compares these values with the best published lower bounds and the known upper bounds.

TABLE 1. Kissing number bounds in dimensions 19 and 21, as of October 2026 [4]. The previous lower bounds are due to Ho [6] in dimension 19 and to Cohn and Li [1] in dimension 21. The upper bounds are the floors of numerical semidefinite programming values [3] and are not interval-certified.

| dimension | previous lower bound | new lower bound | upper bound |
|-----------|----------------------|-----------------|-------------|
| 19        | 11948 [6]            | <b>12268</b>    | 24417 [3]   |
| 21        | 29768 [1]            | <b>30779</b>    | 53524 [3]   |

The mechanism is the slack between 6 and  $\frac{1}{2}\sqrt{8n}$ . Cohn and Li's added points are flat: each is a sign vector scaled to squared norm 8, and none of them touches an old point. We spend this unused room by changing the shape of the added points.

- In 19 dimensions (Section 3) the added points come from 20 dimensions. Projecting them into a hyperplane leaves a short three-dimensional tail, and we multiply it by 9/11 or 7/5 according to its direction. This lets two added points differ in only four head coordinates when their tails differ, which a code of minimum distance at least 5, such as Ho's, must avoid. We add 1792 points where Ho adds 1280. The long tails come too close to 192 old points, which we delete, for a net gain of  $1792 - 192 - 1280 = 320$ .
- In 21 dimensions (Section 4) each added point carries its own positive weight on each coordinate. That freedom allows sign patterns from the even cosets of a  $[21, 9, 6]$  code as well as from the odd ones, where Cohn and Li use only the four odd cosets. The added points are 192 integer vectors and their images under a group of 16 sign changes. The 192 vectors are half of an earlier table; the other half is replaced by their images under one extra sign change that fixes the octad where deletions happen, which cuts the deletions from 29 to 13. The weights overshoot that octad, and we delete 13 points on it, for a net gain of  $3072 - 13 - 2048 = 1011$ .

The two proofs differ in character. The 19-dimensional proof is a case analysis that can be checked by hand, given the finite facts about a code of size 4096 stated in Lemma 3.1. The 21-dimensional proof depends on computer calculation in an essential way, unlike the proofs in [1]. Its added points are integer vectors with entries between  $1.5 \cdot 10^8$  and  $2.5 \cdot 10^8$  in absolute value. They were obtained by numerical optimization and rounding, not by hand. The theorem is a statement about that explicit table, and its proof is a list of about 5.6 million exact integer inequalities. One program checks these in an equivalent form; a second, written independently, checks all 473658031 pairs without the reduction, and we also formalized both results in Lean 4 with Mathlib (Section 5). The formal proof does not verify our case analysis. It verifies the conclusion, that explicit sets of 12268 and 30779 unit vectors with pairwise inner products at most  $1/2$  exist.

## 2. THE ODD-SIGN CONFIGURATIONS

We identify a binary word with its support. For  $c \in \mathbb{F}_2^n$  we write  $\text{wt}(c)$  for its weight,  $(-1)^c \in \{\pm 1\}^n$  for its sign vector, and  $\lambda \circ q$  for the coordinatewise product of two vectors.

Let  $\mathcal{O}$  be a set of words of length  $n$  with weight 8 and pairwise intersections of size at most 4. The *odd configuration*  $B(\mathcal{O})$  consists of the  $4\binom{n}{2}$  roots  $\pm 2e_i \pm 2e_j$  and, for each  $o \in \mathcal{O}$ , the 128 vectors with entries  $\pm 1$  on  $o$ , zeros elsewhere and an odd number of minus signs [1, §2]. All have squared norm 8.

**Lemma 2.1** (Cohn and Li). *Distinct  $b, b' \in B(\mathcal{O})$  satisfy  $\langle b, b' \rangle \leq 4$ .*

*Proof.* Two distinct roots have inner product in  $\{0, \pm 4, -8\}$ . A root has two nonzero coordinates, each contributing at most 2 against an octad vector, so that inner product is at most 4. Two vectors on the same octad have odd sign patterns, so they differ in an even and positive number  $d \geq 2$  of places and have inner product  $8 - 2d \leq 4$ . Two vectors on different octads share at most four coordinates.  $\square$

The next lemma is the reason for odd signs. It is the weighted form of the argument in [1, §2].

**Lemma 2.2.** *Let  $c \in \mathbb{F}_2^n$  meet every word of  $\mathcal{O}$  in an even number of coordinates, let  $q = (-1)^c$ , and let  $\lambda \in \mathbb{R}_{>0}^n$ . Let  $b$  be an octad vector on  $o \in \mathcal{O}$ . Then  $q$  and  $b$  disagree in an odd number of coordinates of  $o$ , and*

$$\langle \lambda \circ q, b \rangle \leq \sum_{j \in o} \lambda_j - 2 \min_{j \in o} \lambda_j,$$

*with equality for some  $b$  on  $o$ . In particular, for  $\lambda = \mathbf{1}$  we have  $\langle q, b \rangle = 6$  if  $q$  and  $b$  disagree in exactly one coordinate and  $\langle q, b \rangle \leq 2$  otherwise. For a root  $b = \pm 2e_i \pm 2e_j$  the inner product is at most  $2(\lambda_i + \lambda_j)$ .*

*Proof.* The sign vector  $q$  has an even number of minus signs on  $o$ , and  $b$  has an odd number. So they disagree in an odd number of coordinates of  $o$ , in particular in at least one, and the bound follows. Equality holds for the odd pattern that agrees with  $q$  except at one coordinate where  $\lambda$  is smallest. For  $\lambda = \mathbf{1}$ ,  $d$  disagreements give  $\langle q, b \rangle = 8 - 2d$  with  $d$  odd.  $\square$

Throughout we test compatibility in integers. For nonzero  $x, y \in \mathbb{Z}^n$  the angle between them is at least  $60^\circ$  if and only if

$$(1) \quad \langle x, y \rangle \leq 0 \quad \text{or} \quad 4\langle x, y \rangle^2 \leq |x|^2 |y|^2.$$

We call such a pair *compatible*; the same test applies to rational vectors after scaling. If every pair from a finite set of nonzero rays is compatible, then the rays normalized to unit length form a kissing configuration, and distinct rays give distinct points because (1) excludes cosine 1. With  $\lambda = \mathbf{1}$ , Lemma 2.2 shows that the sign vector of a word that meets every word of  $\mathcal{O}$  evenly is compatible with  $B(\mathcal{O})$  when  $6 \leq \frac{1}{2}\sqrt{8n}$ , that is, for  $n \geq 18$ , with contacts when  $n = 18$ .

### 3. DIMENSION 19

**3.1. The codes.** Coordinates are  $1, \dots, 20$ . Let  $E \subseteq \mathbb{F}_2^{20}$  be the span of the twelve words

$$\begin{aligned} &\{1, 8, 9, 12, 16, 17, 18, 19\}, \{2, 10, 11, 14, 15, 17, 18, 20\}, \{3, 7, 9, 13, 15, 16, 17, 19\}, \\ &\{4, 7, 8, 10, 12, 15, 16, 19\}, \{5, 10, 12, 13, 15, 16, 17, 18\}, \{6, 7, 8, 9, 10, 13, 16, 18\}, \\ &\{1, 4, 7, 9\}, \{1, 5, 6, 18, 20\}, \{1, 3, 12, 15, 20\}, \{1, 10, 13, 19, 20\}, \\ &\{1, 3, 5, 6, 7, 13, 14, 15, 18, 20\}, \{2, 4, 6, 7, 8, 13, 14, 16, 17, 18\}. \end{aligned}$$

On the first 19 coordinates these are Ho's generators  $m_1, \dots, m_6, s_1, \dots, s_4, r_1, r_2$  [6, §2]. Our coordinate 20 is Ho's coordinate 24, the last column of his extension matrix  $P$ , so we have added it to the words where that column has a 1. Ho shows that the full extension to length 24 is the extended binary Golay code [6, Prop. 2]. So  $E$  is the Golay code punctured at Ho's coordinates 20 to 23, and its dual  $E^\perp$  is the Golay code shortened at the same four coordinates [9]. The 130 words of weight 8 in  $E^\perp$  are the *octads*. They form the shortened Steiner system  $S(5, 8, 24)$  with  $n = 20$  of [1, Table 2.1], so  $B(\text{octads})$  is Cohn and Li's 17400-point configuration  $C_{20}$  in  $\mathbb{R}^{20}$ .

The five octads of the Golay code that contain four given coordinates partition the other twenty coordinates into five tetrads, and these six tetrads form a *sextet* [2, Ch. 10]. Puncturing at the four given coordinates turns those five octads into five words of weight 4 in  $E$ .

The following facts are finite computations on the 4096 words of  $E$ .

**Lemma 3.1.** (1) *The weight enumerator of  $E$  is*

$$\begin{aligned} &1 + 5z^4 + 64z^5 + 240z^6 + 320z^7 + 250z^8 + 640z^9 + 1056z^{10} \\ &+ 640z^{11} + 250z^{12} + 320z^{13} + 240z^{14} + 64z^{15} + 5z^{16} + z^{20}. \end{aligned}$$

The five words of weight 4 are pairwise disjoint and cover all 20 coordinates.

- (2) Let  $T = \{1, 4, 7, 9\}$ , one of the five, and let  $H$  be the other 16 coordinates, the head. For  $e \in E$  write  $e_T$  and  $e_H$  for its parts on  $T$  and  $H$ . The nonzero words with  $\text{wt}(e) \leq 7$  have  $(\text{wt}(e_T), \text{wt}(e_H))$  equal to  $(4, 0)$  for 1 word,  $(0, 4)$  for 4,  $(1, 4)$  for 64,  $(0, 6)$  for 96,  $(2, 4)$  for 144,  $(1, 6)$  for 256 and  $(3, 4)$  for 64.
- (3) Two distinct octads meet in at most 4 coordinates. The octads meet  $T$  in 0, 2 or 4 coordinates, for 30, 96 and 4 octads respectively.

The intersections in (3) are even because  $T \in E$  and the octads lie in  $E^\perp$ . By (2), two distinct words  $c, c' \in E$  with  $d_T = \text{wt}(c_T + c'_T)$  and  $d_H = \text{wt}(c_H + c'_H)$  satisfy the following. If  $d_T = 0$ , then  $d_H = 4$  (and  $c + c'$  is a tetrad inside  $H$ ) or  $d_H \geq 6$ . If  $1 \leq d_T \leq 3$ , then  $d_H \geq 4$ . If  $d_T = 4$ , then  $d_H = 0$  (and  $c' = c + \mathbf{1}_T$ ),  $d_H = 4$  or  $d_H \geq 6$ ; this is the case  $d_T = 0$  applied to  $c$  and  $c' + \mathbf{1}_T \in E$ .

**3.2. The configuration.** Let

$$R = \begin{pmatrix} 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix},$$

acting on the coordinates  $(1, 4, 7, 9)$  of  $T$ . The rows of  $R/2$  are an orthonormal basis of  $\mathbf{1}_T^\perp \cap \mathbb{R}^T$ , so  $(R/2)^\top(R/2) = I - J/4$ . The map  $\varphi(v) = (v_H, Rv_T/2) \in \mathbb{R}^{16} \times \mathbb{R}^3$  therefore satisfies

$$(2) \quad \langle \varphi(u), \varphi(v) \rangle = \langle u, v \rangle - \frac{1}{4} \left( \sum_{j \in T} u_j \right) \left( \sum_{j \in T} v_j \right),$$

and it is an isometry from the hyperplane  $\mathbf{1}_T^\perp$  onto  $\mathbb{R}^{19}$ . The configuration in  $\mathbb{R}^{19}$  has two parts.

*Base.* There are 10668 points of  $C_{20}$  orthogonal to  $\mathbf{1}_T$ , as many as in Leech's 19-dimensional configuration. We delete from them the 192 points that lie on the four octads containing  $T$ ; these are the points with two plus and two minus signs on  $T$ . The remaining 10476 points  $b$  enter as  $\varphi(b)$ . They are the 480 roots inside  $H$ , the 12 roots  $\pm 2(e_i - e_j)$  with  $i, j \in T$ , the 3840 odd vectors on the 30 octads disjoint from  $T$ , and the 6144 odd vectors on the 96 octads meeting  $T$  twice whose two  $T$ -entries have opposite signs.

*Additions.* Let  $c \in E$  with  $\text{wt}(c_T) \in \{1, 2, 3\}$  and  $q = (-1)^c$ . The projected tail  $t_0 = Rq_T/2$  satisfies  $|t_0|^2 = 4 - \frac{1}{4}(\sum_T q_j)^2$  by (2). When  $\text{wt}(c_T)$  is odd,  $t_0 \in \{\pm 1\}^3$  and  $|t_0|^2 = 3$ ; we call  $c$  *diagonal*. When  $\text{wt}(c_T) = 2$ ,  $t_0 = \pm 2e_k$  and  $|t_0|^2 = 4$ ; we call  $c$  an *axis* word. The added point is

$$a(c) = (q_H, \alpha t_0), \quad \alpha = \begin{cases} 9/11, & c \text{ diagonal,} \\ 7/5, & c \text{ axis,} \end{cases}$$

with  $|a(c)|^2 = 2179/121$  or  $596/25$ . Multiplying by 22 or 10 makes it an integer vector, and the base points  $\varphi(b)$  become integer vectors after multiplying by 2.

*Which words.* Let  $W$  be the span of the five tetrads, a code of size 32. The 3584 words with  $\text{wt}(c_T) \in \{1, 2, 3\}$  fall into 112 cosets of  $W$ . Each coset  $c + W$  is a 5-cube: its vertices are the 32 words, and each of the five directions adds one tetrad. We keep one of the two bipartition classes of each cube, 16 words. Lemma 3.2 below holds for either class, so any choice gives a valid configuration of the same size. For definiteness, and to match `data/D19_12268_rays.txt`, we keep the words  $c$  for which  $c + c_{\min}$  is a sum of an even number of tetrads, where  $c_{\min}$  is the smallest word of the coset read as the integer  $\sum_{j \in c} 2^{j-1}$ . This gives 1024 diagonal and 768 axis words, 1792 in all.

The total is  $10668 - 192 + 1792 = 12268$ . Table 2 lists the blocks.

TABLE 2. The blocks of the 19-dimensional configuration.

| block                                  | size  | squared norm |
|----------------------------------------|-------|--------------|
| roots in $H$                           | 480   | 8            |
| roots in $T$                           | 12    | 8            |
| octad vectors, octad disjoint from $T$ | 3840  | 8            |
| octad vectors, octad meeting $T$ twice | 6144  | 8            |
| diagonal additions                     | 1024  | 2179/121     |
| axis additions                         | 768   | 596/25       |
| total                                  | 12268 |              |

TABLE 3. Largest inner product between a base point and an addition in dimension 19.

| base points              | count | reason                                         | diagonal | axis   |
|--------------------------|-------|------------------------------------------------|----------|--------|
| roots in $H$             | 480   | $b_T = 0$                                      | 4        | 4      |
| roots in $T$             | 12    | $\alpha \langle q_T, b_T \rangle \leq 4\alpha$ | 36/11    | 28/5   |
| octads disjoint from $T$ | 3840  | $b_T = 0$ , Lemma 2.2                          | 6        | 6      |
| octads meeting $T$ twice | 6144  | three cases above                              | 6        | 34/5   |
| bound $\sqrt{2 a ^2}$    |       |                                                | 6.0014   | 6.9051 |

**3.3. Proof.** Some restriction inside each cube is needed, because two words of one cube that differ by a tetrad inside  $H$  have the same  $T$ -pattern and  $d_H = 4$ . For two diagonal words this gives inner product  $8 + 3(9/11)^2 \approx 10.01$  against the bound  $\frac{1}{2}|a|^2 \approx 9.00$ .

**Lemma 3.2.** *Two distinct kept words  $c, c'$  satisfy  $c_H \neq c'_H$ , and  $d_H \geq 6$  if  $c_T = c'_T$ .*

*Proof.* If  $c_H = c'_H$ , then  $c + c'$  is a word supported in  $T$ , so it is  $\mathbf{1}_T$  by Lemma 3.1(2). Then  $c$  and  $c'$  lie in the same coset of  $W$  and differ by one tetrad, so they lie in opposite classes of the 5-cube and are not both kept. If  $c_T = c'_T$  and  $d_H = 4$ , then  $c + c'$  is a tetrad in  $H$ , and the same argument applies.  $\square$

We now check every pair, using the unnormalized vectors.

*Base with base.* By (2) the map  $\varphi$  preserves inner products on  $\mathbf{1}_T^\perp$ , so Lemma 2.1 gives  $\langle \varphi(b), \varphi(b') \rangle \leq 4 = \frac{1}{2} \cdot 8$ .

*Base with additions.* Since  $\sum_T b_j = 0$ , equation (2) gives

$$(3) \quad \langle a(c), \varphi(b) \rangle = \langle q_H, b_H \rangle + \alpha \langle q_T, b_T \rangle = \langle q, b \rangle + (\alpha - 1) \langle q_T, b_T \rangle.$$

The word  $c$  lies in  $E$ , so it meets every octad evenly, and Lemma 2.2 applies: for an octad vector  $b$ ,  $\langle q, b \rangle = 6$  if  $q$  and  $b$  disagree in exactly one place of the octad and  $\langle q, b \rangle \leq 2$  otherwise. The bound to meet is  $\frac{1}{2}|a|\sqrt{8} = \sqrt{2|a|^2}$ , which is  $\sqrt{4358/121} \approx 6.0014$  for diagonal and  $\sqrt{1192/25} \approx 6.9051$  for axis additions.

For the 6144 points on octads meeting  $T$  twice,  $b_T = \pm(e_i - e_j)$  with  $i, j$  the two coordinates of the octad in  $T$ , so  $\langle q_T, b_T \rangle \in \{-2, 0, 2\}$ . If it is 0, the inner product is  $\langle q, b \rangle \leq 6$ . If it is 2, the inner product is at most  $6 + 2(\alpha - 1)$ , which is  $62/11$  or  $34/5$ . If it is  $-2$ , then  $q$  and  $b$  disagree at both coordinates of the octad in  $T$ . The number of disagreements is odd, so it is at least 3,  $\langle q, b \rangle \leq 2$ , and the inner product is at most  $2 - 2(\alpha - 1)$ . Table 3 collects the maxima.

TABLE 4. Largest inner products between two additions in dimension 19, for the cases with nonnegative tail product. The bound is  $\frac{1}{2}|a||a'|$ .

| types              | case                                              | $\langle a, a' \rangle$ | bound                | cosine |
|--------------------|---------------------------------------------------|-------------------------|----------------------|--------|
| diagonal, diagonal | $d_T = 0, d_H = 6$                                | 727/121                 | 2179/242             | 0.3337 |
|                    | $d_T = 2, \langle t_0, t'_0 \rangle = 1, d_H = 4$ | 1049/121                | 2179/242             | 0.4814 |
| axis, axis         | $d_T = 0, d_H = 6$                                | 296/25                  | 298/25               | 0.4966 |
|                    | $d_T = 2, d_H = 4$                                | 8                       | 298/25               | 0.3356 |
| diagonal, axis     | $d_T = 1, d_H = 4$                                | 566/55                  | $\sqrt{324671/3025}$ | 0.4967 |

All maxima are below the bounds, since  $36 \cdot 121 = 4356 < 4358$  and  $(34/5)^2 = 1156/25 < 1192/25$ .

*Additions with additions.* For kept words  $c \neq c'$ ,

$$\langle a(c), a(c') \rangle = 16 - 2d_H + \alpha\alpha' \langle t_0, t'_0 \rangle, \quad \langle t_0, t'_0 \rangle = \langle q_T, q'_T \rangle - \frac{1}{4} \left( \sum_T q_j \right) \left( \sum_T q'_j \right).$$

Here  $d_H \geq 4$  in every case, and  $d_H \geq 6$  when  $c_T = c'_T$ , by Lemma 3.1(2) and Lemma 3.2. The tail product depends only on the two patterns on  $T$ :

- two diagonal words have  $d_T \in \{0, 2, 4\}$  and  $\langle t_0, t'_0 \rangle = 3, \pm 1, -3$  respectively;
- two axis words have  $d_T \in \{0, 2, 4\}$  and  $\langle t_0, t'_0 \rangle = 4, 0, -4$  respectively;
- a diagonal and an axis word have  $d_T \in \{1, 3\}$  and  $\langle t_0, t'_0 \rangle = 2$  or  $-2$  respectively.

The cases with  $\langle t_0, t'_0 \rangle < 0$  give at most 8, below every bound. Table 4 lists the others at the smallest allowed  $d_H$ .

Every entry is strictly below its bound. For the diagonal–axis row we compare  $(566/55)^2 = 320356/3025$  with  $324671/3025$ . So all pairs are compatible. The roots inside  $H$  and inside  $T$  already span  $\mathbb{R}^{19}$ . This proves Theorem 1.1 for  $n = 19$ .

**3.4. Remarks.** The tail lengths have little room. A diagonal addition against an octad vector on an octad disjoint from  $T$  needs  $36 \leq 2(16 + 3\alpha^2)$ , so  $\alpha \geq \sqrt{2/3} \approx 0.8165$  for diagonal words. Two axis additions with the same pattern at  $d_H = 6$  need  $4 + 4\alpha^2 \leq \frac{1}{2}(16 + 4\alpha^2)$ , so  $\alpha \leq \sqrt{2} \approx 1.4142$  for axis words. The diagonal–axis pairs at  $d_T = 1, d_H = 4$  couple the two factors. Numerically, the conditions behind Tables 3 and 4 allow the diagonal factor in about  $[0.8165, 0.877]$  and the axis factor in about  $[1.304, 1.4142]$ , in a thin region where a larger axis factor permits a larger diagonal factor. We chose  $9/11$  and  $7/5$  as simple fractions inside it. With plain projection, both factors equal to 1, the diagonal–axis pairs have inner product 10 against  $\frac{1}{2}\sqrt{19 \cdot 20} \approx 9.75$ , so plain projection fails.

Axis factors above 1 are what the 192 deleted points cannot tolerate. A deleted point has  $b_T \in \{\pm 1\}^4$  with two signs of each kind. An axis word with  $q_T = b_T$  whose sign vector disagrees with  $b$  in exactly one place of the octad gives inner product  $6 + 4 \cdot \frac{2}{5} = 38/5 > 6.9051$ . A computer check shows that each deleted point conflicts in this way with some kept axis addition.

We have not determined the largest configuration of this shape. One numerical observation bears on it. The 3584 candidate words have exactly 1792 distinct heads, so both bipartition classes use the same heads. A numerical search over all  $2^{16}$  sign vectors on  $H$ , looking for a tail compatible with this 10476-point base by local optimization from 48 starting points, found one only for these 1792 heads. A local search can miss a feasible tail, so this is not a proof.

The additions touch nothing. By direct computation, all 2386584 contacts (pairs with cosine exactly  $1/2$ ) are between base points. The closest pairs involving an addition have cosine  $\sqrt{1089/4358} \approx 0.49989$ , a diagonal addition against an octad vector on an octad disjoint from  $T$ .

#### 4. DIMENSION 21

**4.1. The codes.** Coordinates are  $1, \dots, 24$ , and the construction lives on the first 21. Let  $G$  be the extended binary Golay code: the cyclic code of length 23 generated by  $g(t) = 1 + t + t^5 + t^6 + t^7 + t^9 + t^{11}$ , with the coefficient of  $t^k$  at coordinate  $k + 1$ , extended by an overall parity bit at coordinate 24 [2, Ch. 3, §2.8]. Let  $\mathcal{O}$  be the 210 octads of  $G$  that vanish on coordinates 22, 23 and 24. Then  $B = B(\mathcal{O})$  is Cohn and Li's 27720-point configuration in  $\mathbb{R}^{21}$ .

Let  $D$  be  $G$  punctured to the first 21 coordinates, a  $[21, 12, 5]$  code. Because  $G$  is self-dual and the octads in  $\mathcal{O}$  vanish on the punctured coordinates, every word of  $D$  meets every octad of  $\mathcal{O}$  evenly. We call the 21 words of weight 5 in  $D$  *lines*. Two sign vectors of words of  $D$  at distance  $d$  have cosine  $1 - 2d/21$ , which exceeds  $1/2$  only for  $d = 5$ , where it is  $11/21$ . Cohn and Li use the 2048 odd-weight words of  $D$ . The sum of two odd words is even, so no two of them are at distance 5.

Let  $C \subseteq D$  be the span of the nine words (bit  $j$  of the binary expansion is coordinate  $j + 1$ )

1559917, 15163, 774651, 1540761, 485602, 408336, 678478, 1748752, 1231514.

It is a  $[21, 9, 6]$  code with only even weights, and  $D/C$  has eight cosets. Four are even, with representatives 0, 1393, 2139, 3370, and contain no lines. Four are odd, with representatives 696, 1993, 2787, 3986, and contain 6, 3, 6 and 6 lines. These are finite computations on the 4096 words of  $D$ . The code  $C$  first appeared in an unpublished 30280-point certificate produced by an AI agent outside this project.

The following lemma is not used in the proof. It explains why the new points must be weighted.

**Lemma 4.1.** *Let  $c, c'$  be distinct words of  $D$ . If they lie in the same coset of  $C$ , or in two cosets of the same parity, then  $\text{wt}(c + c') \geq 6$ . If they lie in cosets of opposite parity, then  $c + c'$  ranges over a whole odd coset as  $c$  and  $c'$  range over their cosets, so  $\text{wt}(c + c') = 5$  occurs.*

*Proof.* In the first case  $c + c'$  lies in  $C$  or in an even coset, and the even words of  $D$  have weight at least 6. In the second case  $c + c'$  lies in an odd coset, and every odd coset contains a line.  $\square$

So flat sign vectors from an even coset and an odd coset conflict at cosine  $11/21$ .

**4.2. Weighted sign vectors.** Here is the simplest way around the conflict. Let  $\ell$  be a line, and put  $\lambda_j = \sqrt{2/5}$  for  $j \in \ell$  and  $\lambda_j = 1$  otherwise, so  $|\lambda|^2 = 18$ . For  $c \in D$  the point  $\lambda \circ (-1)^c$  has, by Lemma 2.2, inner product at most  $\sum_{j \in \mathcal{O}} \lambda_j - 2 \min_{j \in \mathcal{O}} \lambda_j$  with the octad vectors on  $\mathcal{O}$ . An octad meets  $\ell$  in 0, 2 or 4 coordinates, because  $\ell \in D$ , and the bound is then  $8 - 2 = 6$ ,  $6 + 2\sqrt{2/5} - 2\sqrt{2/5} = 6$ , or  $4 + 2\sqrt{2/5} < 6$ . This is at most  $\frac{1}{2}\sqrt{8}|\lambda| = 6$ . So among weight vectors equal to 1 off a line and constant on it,  $\sqrt{2/5}$  is the smallest weight on the line that the base allows. A small weight on  $\ell$  weakens the conflicts at distance 5 that run through  $\ell$ .

In general, let  $c \in D$ ,  $q = (-1)^c$  and  $\lambda \in \mathbb{R}_{>0}^{21}$ . Lemma 2.2 shows that  $\lambda \circ q$  is compatible with every point of  $B$  exactly when

$$(4) \quad \sum_{j \in \mathcal{O}} \lambda_j - 2 \min_{j \in \mathcal{O}} \lambda_j \leq \sqrt{2}|\lambda| \quad \text{for all } \mathcal{O} \in \mathcal{O}, \quad \lambda_i + \lambda_j \leq \frac{1}{\sqrt{2}}|\lambda| \quad \text{for all } i \neq j.$$

The hypothesis  $c \in D$  is what makes the first condition exact. Two points  $\lambda \circ q$  and  $\mu \circ q'$  have inner product  $\sum_j \lambda_j \mu_j (-1)^{(c+c')_j}$ , which depends only on the weights and on  $c + c'$ .

**4.3. The configuration.** *Base.* Let  $Q = \{3, 6, 10, 11, 16, 18, 20, 21\} \in \mathcal{O}$ . Delete from  $B$  the 13 odd sign patterns on  $Q$  whose signs, in the coordinate order of  $Q$ , are

$$\begin{aligned} & +---+----+, +---+---+-, +-+---+---, +-+---++++, +-+++++---, \\ & +-+++++---, ++---++---, ++---++++-, ++-++++---, +++-+++++, \\ & +++-+-+---, ++++++---, +++++-+++. \end{aligned}$$

The certificate file lists them in this order as `removed_base_representatives`. We know no closed rule for them; they are exactly the odd patterns on  $Q$  that have cosine above  $1/2$  with some new point (Section 4.5). The rest,  $B'$ , has  $840 + 209 \cdot 128 + 115 = 27707$  points.

*Sign group.* Let  $\Sigma \subseteq D$  be the span of 92361, 20689, 346115 and  $w = 14723$ , with supports  $\{1, 4, 7, 8, 12, 14, 15, 17\}$ ,  $\{1, 5, 7, 8, 13, 15\}$ ,  $\{1, 2, 12, 15, 17, 19\}$ ,  $\{1, 2, 8, 9, 12, 13, 14\}$ .

For  $h \in \Sigma$  let  $S_h$  change the signs of the coordinates in  $h$ . The group  $\{S_h\}$  has order 16. The first three generators span a subgroup  $\Sigma_0 \subseteq C$  of order 8; the word  $w$  has odd weight, so  $w \notin C$ , but  $w \in D$ . Section 4.5 explains where  $w$  comes from. All four generators miss  $Q$ , so every  $S_h$  fixes every vector supported on  $Q$ , including the 13 deleted points.

*New points.* Let  $r_1, \dots, r_{192} \in \mathbb{Z}^{21}$  be the first 192 vectors in the `representatives` field of `data/D21_30779_certificate.json` (see Data). (That field has 384 entries; the last 192 are  $S_w r_1, \dots, S_w r_{192}$ , so that the file also describes the configuration as 384 representatives under  $\Sigma_0$ . Both readings give the same set of points.) They have no zero entries, their entries have absolute values between 149071270 and 249507763, and their squared norms lie between  $0.99999999 \cdot 10^{18}$  and  $1.000000002 \cdot 10^{18}$ . Appendix A gives one of them for each coset below. The new points are the  $192 \cdot 16 = 3072$  vectors  $S_h(r_i)$ .

The sign words of  $r_1, \dots, r_{192}$  and their  $\Sigma_0$ -images (the words  $c$  with  $\text{sign}(r) = (-1)^c$ ), 1536 words in all, fill half of each of the six cosets  $c_0 + C$  with

$$c_0 \in \{696, 1993, 2787\} \text{ (odd)}, \quad c_0 \in \{1393, 2139, 3370\} \text{ (even)},$$

with 32 representatives and 256 words on each. Adding  $w$  swaps the cosets in the pairs  $0 \leftrightarrow 696$ ,  $1393 \leftrightarrow 1993$ ,  $2139 \leftrightarrow 2787$  and  $3370 \leftrightarrow 3986$ . So the 3072 sign words of all new points meet every coset of  $C$ : 512 words on each of 1393, 1993, 2139, 2787 and 256 on each of 0, 696, 3370, 3986. Cohn and Li's four odd cosets are all present, together with all four even ones. On the cosets of 1993 and 2139 the weights of  $r_i$  follow the line model, with the lines  $\{8, 15, 16, 18, 19\}$  and  $\{1, 3, 5, 17, 18\}$ : after scaling each representative to  $|\lambda|^2 = 18$ , every entry is within  $10^{-6}$  of the model. On the coset of 3370 they follow it only approximately, with the line  $\{4, 5, 6, 8, 10\}$  and every entry within  $7.6 \cdot 10^{-3}$ . The cosets of 2139 and 3370 are even and contain no line. Their lines lie in the odd cosets of 1993 and 696, and adding the line to a word of either even coset gives a word in the coset of 3986, at distance 5. So the reduced weights weaken exactly the conflicts with the coset of 3986. On the other three cosets the weights are irregular, and each representative's entries lie between 0.82 and 1.15 times their root mean square. The weights also vary slightly between the 32 orbits of  $\Sigma_0$  inside a coset.

We do not have a closed form for the irregular weights. They were found by numerical optimization of the conditions of the previous subsection, with a target cosine slightly below  $1/2$ , and then rounded to integers at scale about  $2 \cdot 10^8$ . The proof does not depend on how they were found. Replacing the three line-model cosets by their exact limits would not shorten the proof, because the irregular cosets would still need a computer check.

The total is  $27707 + 3072 = 30779$ .

#### 4.4. Proof.

**Lemma 4.2.** *Every  $S_h$ ,  $h \in \Sigma$ , maps  $B'$  onto itself.*

*Proof.* Every word of  $\Sigma$  lies in  $D$ , so it meets every octad of  $\mathcal{O}$  in an even number of coordinates. (Membership of the four generators in  $D$  is a finite check: the first three lie in  $C \subseteq D$ , and  $w$  lies in  $D$  but not in  $C$ .) A sign change  $S_h$  maps roots to roots. On an octad vector with support  $o$  it changes an even number of signs, so the number of minus signs on  $o$  stays odd. So  $S_h(B) = B$ . Since  $S_h$  fixes the 13 deleted points,  $S_h(B') = B'$ .  $\square$

Each  $S_h$  is orthogonal and  $S_a S_b = S_{a+b}$ . Hence

$$\langle S_a r_i, S_b r_j \rangle = \langle r_i, S_{a+b} r_j \rangle, \quad \langle S_a r_i, b \rangle = \langle r_i, S_a b \rangle.$$

With Lemma 4.2, every pair involving a new point is determined by one of these finite lists:

- (1) the  $\binom{192}{2} \cdot 16 + 192 \cdot 15 = 296256$  products  $\langle r_i, S_h r_j \rangle$  with  $i \leq j$ , excluding  $i = j$ ,  $h = 0$ ;
- (2) the  $192 \cdot 27707 = 5319744$  products  $\langle r_i, b \rangle$  with  $b \in B'$ .

**Proposition 4.3.** *Let  $L = 2 \cdot 10^8$ . Every product in lists (1) and (2), with  $x, y$  the two vectors, satisfies  $\langle x, y \rangle \leq (\frac{1}{2} - \frac{1}{L})|x||y|$ . In particular every such pair satisfies (1).*

*Proof.* This is a finite computation in exact integers. For  $g = \langle x, y \rangle > 0$  the inequality is equivalent to  $(2Lg)^2 \leq (L-2)^2|x|^2|y|^2$ . For list (2), Lemma 2.2 gives the maximum of  $\langle r_i, b \rangle$  over the full fiber of odd vectors on each octad other than  $Q$  and over the roots in closed form, and the 115 retained points on  $Q$  are checked one by one. The program `scripts/verify_d21_orbits.py` checks this with Python integers in about 2 seconds. It first checks that representatives 193 to 384 of the certificate file are  $S_w r_1, \dots, S_w r_{192}$ , and then runs the equivalent computation for the 384 representatives under  $\Sigma_0$ : 590976 orbit products and the products with  $B'$ . A second program, `scripts/check_pairs.py`, written independently, does not use the reduction and tests all 473658031 pairs of the configuration directly, in about one minute. The Lean formalization of Section 5 also checks all pairs directly.  $\square$

Base pairs are compatible by Lemma 2.1. So all 30779 points are pairwise compatible, and they span  $\mathbb{R}^{21}$  because  $B'$  contains all 840 roots. This proves Theorem 1.1 for  $n = 21$ .

**4.5. Remarks.** The smallest value of  $\frac{1}{4} - \cos^2$  over pairs involving a new point is

$$\frac{105355791297}{7999999998594867688} \approx 1.32 \cdot 10^{-8},$$

attained by a new point and an octad vector on an octad other than  $Q$ . The tightest pair of two new points has  $\frac{1}{4} - \cos^2 \approx 1.98 \cdot 10^{-8}$ ; their sign vectors are at Hamming distance 5. All 12975933 contacts are between base points.

The deletion is necessary for these representatives. A computer check shows that each of the 13 deleted points has cosine between 0.50444 and 0.50496 with some new point, and that each of the other 115 odd patterns on  $Q$  has cosine at most 0.49999998 with every new point. The group  $\{S_h\}$  fixes  $Q$  pointwise, so its orbits are free to carry different weights near  $Q$ .

*Where the second half of the table comes from.* The table came in two steps, and the second step rests on the following observations; the proof above checks the final table and nothing else.

- (a) For every  $h \in D$  that vanishes on  $Q$ , the proof of Lemma 4.2 shows that  $S_h$  maps  $B'$  onto itself and fixes each of the 13 deleted points.
- (b) So if a set  $A$  of new points is compatible with  $B'$  and with itself, then so is  $S_h(A)$ , and  $A \cup S_h(A)$  needs no further deletions. The only new conditions are the pairs between  $A$  and  $S_h(A)$ .

- (c) Exactly 32 words of  $D$  vanish on  $Q$ , and 8 of them form  $\Sigma_0$ . So up to  $\Sigma_0$  there are three choices of  $h$  besides  $\Sigma_0$  itself. For  $A = \{S_g r_i\}$ ,  $g \in \Sigma_0$ , the largest cosine between  $A$  and  $S_h(A)$  is 0.4746 on the class of  $w$ , but 0.5117 and 0.9794 on the other two classes, which therefore fail.
- (d) An earlier configuration of 30763 points used 384 representatives under  $\Sigma_0$  and deleted 29 points on  $Q$ . In it, the graph on the 384 orbits that joins two orbits when some pair between them has cosine above 0.4999 has two components of 192 orbits, and no pair across the components has cosine above 0.491. The 29 deleted patterns split disjointly, 13 conflicting with one component and 16 with the other. Keeping the first component,  $r_1, \dots, r_{192}$ , needs only 13 deletions, and (b) with  $h = w$  refills the other 1536 points. This is the whole gain over 30763.
- (e)  $S_w$  moves the sign words of  $A$  from the cosets of 696, 1993, 2787, 1393, 2139, 3370 to those of 0, 1393, 2139, 1993, 2787, 3986, in that order. This is why all eight cosets appear. By Lemma 4.1, flat sign vectors from cosets of opposite parity conflict at cosine 11/21; with these weights the largest cross cosine is 0.4746.

These facts were observed numerically.

With the base held fixed, each new point may be moved independently to any unit vector within chord distance  $1/(8 \cdot 10^8)$  of its position, and the result is again a kissing configuration, because each inner product then changes by at most  $2/(8 \cdot 10^8) < 1/L$ . So with  $B'$  fixed these configurations form an open set of dimension  $3072 \cdot 20 = 61440$ . This freedom does not seem to produce room for a 30780th point, and we have not tried to prove that it cannot.

## 5. FORMAL VERIFICATION

We formalized Theorem 1.1 in Lean 4 [5] with Mathlib [10], using Lean 4.33.1 and Mathlib at tag `v4.33.1`. The main theorems `kissing_d19` and `kissing_d21` state, for  $(n, N) = (19, 12268)$  and  $(21, 30779)$ , that there is a finite set  $S$  in `EuclideanSpace ℝ (Fin n)` with

$$|S| = N, \quad \|x\| = 1 \quad (x \in S), \quad \langle x, y \rangle \leq \frac{1}{2} \quad (x \neq y \in S).$$

A further lemma restates this in terms of spheres: the unit balls centred at  $2x$ ,  $x \in S$ , touch the unit ball at the origin, and their centres are at pairwise distance at least 2. We do not define  $\tau_n$  in Lean. The theorems are the existence statements that the bounds  $\tau_n \geq N$  mean.

The development has two layers.

- (1) A plain Lean library, `lean/KissingCheck`, builds each configuration as an array of integer rays from its compact description: Ho's twelve generators and the rules of Section 3, or the Golay polynomial, the 13 deleted sign patterns, the four generators of  $\Sigma$  and the 192 representatives. Each ray is a positive multiple of a point in the text; for instance, the 19-dimensional base rays are  $(2b_H, Rb_T)$ . One Boolean function tests (1) on every pair, and another tests that every ray has  $n$  coordinates and is nonzero.
- (2) A Mathlib library, `lean/KissingD19D21` (the bridge is in `Bridge.lean`), proves, with kernel-checked proofs, that if both tests return true then the normalized rays are  $N$  distinct unit vectors with pairwise inner products at most  $1/2$ . It identifies the Euclidean inner product of two rays with their integer dot product, derives cosine at most  $1/2$  from (1), and uses cosine less than 1 to show that the normalized rays are distinct.

The tests are evaluated with `native_decide`. This tactic compiles the Boolean function, runs it, and if the result is true adds an axiom asserting that it evaluates to true. Each use appears in `#print axioms` as its own auxiliary axiom. Each main theorem depends on `propext`, `Classical.choice`, `Quot.sound` and three such axioms, for the size of the array, the nonzero rays and the pairs. Trusting them means trusting that Lean's compiler, its C

code generator and its arbitrary-precision integer runtime compute what the definitions say. A kernel evaluation of 473 million pairs is out of reach. Running both tests natively with `lake exe checkall 19 21` takes about one minute on one core. Once Mathlib is built, the whole project builds in about three minutes on a laptop, about 90 seconds of it for the 21-dimensional module.

Only the conclusion is formally verified. The theorems depend on the three tests and on the bridge in layer (2), not on the construction code. Lean does not verify that the arrays it builds match the descriptions in Sections 3 and 4, and it does not verify our case analysis. We checked the match separately. The script `scripts/compare_with_witness.py` reduces each ray built in Lean (written to `lean/out/` by `lake exe dumprays`) and each ray in the files `data/D19_12268_rays.txt` and `data/D21_30779_rays.txt` to its primitive integer vector, and finds the two sets equal, with 12268 and 30779 directions.

The Lean code names some objects differently from this paper: there  $K$  is  $\Sigma$ ,  $H$  is  $\Sigma_0$  and  $T$  is the 21-dimensional octad  $Q$ .

The development also contains sanity checks. Kernel-checked `decide` proofs show that the pair test accepts cosine exactly  $1/2$  and rejects a pair with  $4\langle x, y \rangle^2 - |x|^2|y|^2 = 1$  at size  $4 \cdot 10^{18}$ . Further checks in `lean/KissingD19D21/Controls.lean`, again by `native.decide`, show that each of the 192 and 13 deleted points conflicts with some added point.

## 6. CONCLUSION

We have shown that  $\tau_{19} \geq 12268$  and  $\tau_{21} \geq 30779$ , improving the best published lower bounds of 11948 and 29768. Two questions about the 21-dimensional construction are open to us. Can some of the 13 deleted points be restored by moving the 192 representatives? And does the step of Section 4.5, keeping one of two decoupled halves of a table and refilling the other with a sign change that fixes the deleted octad, work in 20 or 22 dimensions?

## USE OF ARTIFICIAL INTELLIGENCE

GPT-6-Astra Pro and Claude Opus 5.5 were used to find these constructions, write the Lean formalization, and draft this paper.

## DATA

The accompanying GitHub repository is

<https://github.com/Felpix-Studios/D19-D21-Kissing>.

It contains short plain-text descriptions of both constructions (`constructions/`), the coordinates of both configurations and the certificate file with the 384 representatives of Section 4 (`data/`), the verification programs (`scripts/`) and the Lean project (`lean/`).

## APPENDIX A. SAMPLE REPRESENTATIVES

Table 5 lists the first representative on each of the six cosets met by  $r_1, \dots, r_{192}$ , in the order in which they appear in `data/D21_30779_certificate.json`. Each is written as three rows of seven entries, coordinates 1 to 21.

TABLE 5. One representative  $r_i$  per coset of  $C$ .

|                        |            |            |            |            |            |            |            |
|------------------------|------------|------------|------------|------------|------------|------------|------------|
| $r_1$ , coset 696      | 246234381  | 219980925  | 209785187  | −202482891 | −224617629 | −218092629 | 216543425  |
|                        | −219980925 | 217704730  | −239855454 | 239957324  | 216009092  | 220121191  | 204061050  |
|                        | 236622028  | 201715279  | 231423123  | 216136122  | 224959675  | 180951525  | 181953903  |
| $r_{33}$ , coset 1993  | 235702242  | −235702246 | 235702245  | 235702249  | −235702248 | −235702244 | −235702246 |
|                        | −149071276 | 235702243  | 235702244  | −235702242 | −235702243 | −235702244 | −235702248 |
|                        | 149071272  | 149071276  | 235702246  | 149071274  | 149071274  | 235702248  | 235702248  |
| $r_{65}$ , coset 2787  | 225351549  | 182591213  | 247281647  | −228539751 | −187676135 | 235642930  | −216731283 |
|                        | −207740975 | −197342048 | 231905242  | 225679867  | 188757907  | −222180761 | −210430962 |
|                        | 204605411  | 249283059  | 215561908  | 215633807  | 221449791  | 227383590  | 225778340  |
| $r_{97}$ , coset 1393  | 214926831  | −190063656 | 233096365  | −230585945 | 200371893  | 241359405  | −235160970 |
|                        | 205103483  | 182738919  | −215303842 | −210489005 | −184471317 | −226272166 | −204197249 |
|                        | 239084407  | 234925920  | 224054811  | 221607207  | 221607197  | 229029592  | 224055305  |
| $r_{129}$ , coset 2139 | 149071274  | 235702247  | 149071275  | 235702251  | 149071276  | −235702243 | −235702244 |
|                        | 235702247  | −235702244 | −235702243 | 235702242  | 235702244  | −235702243 | −235702248 |
|                        | 235702242  | 235702248  | 149071275  | 149071274  | 235702242  | 235702247  | 235702249  |
| $r_{161}$ , coset 3370 | 235368146  | −235367792 | 235367792  | −149956790 | 149584899  | −150128199 | 235368272  |
|                        | 149956667  | −235221677 | 150184393  | −235795937 | −235221410 | 235740042  | 235392962  |
|                        | 235917546  | 235399007  | 235796162  | 235739740  | 235942608  | 235368197  | 235399408  |

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